HSC Extension 2 Mathematics

401

Mechanics

★

A particle is describing SHM in a straight line with an amplitude of 2 metres. Its speed is 3m/s when the particle is 1 metre from the centre of motion.

What is the period of the motion?

A) $\frac{\pi\sqrt{3}}{2}$ B) $\frac{2\pi\sqrt{3}}{3}$ C) $\pi\sqrt{3}$ D) $\frac{2\pi\sqrt{2}}{3}$

HSC Extension 2 Mathematics

402

Mechanics

★★★

The velocity v m/s of a particle moving in simple harmonic motion along the x axis is given by $v^2 = 8 - 2x - x^2$.

(i) Between which two points is the particle oscillating? 2

(ii) Find the acceleration of the particle in terms of x 1

(iii) Find the period of the motion 1

(iv) What is the time taken by the particle to travel the first 60m of its motion (leave in exact form) 1

HSC Extension 2 Mathematics

403

Mechanics

★★

The rise and fall of tides can be approximated to simple harmonic motion. At 11am the depth of water in a tidal lagoon is lowest at 1m. The following high tide is at 5:21pm with a depth of 5m.

(i) Find the period of this motion in minutes 1

(ii) Express the motion using a cosine function 2

(iii) Calculate between which times a yacht could safely cross the lagoon if a minimum depth of 2m is required 2

HSC Extension 2 Mathematics

404

Mechanics

★★★

A particle moves along the $x-$ axis, starting at $x = 0.1$ at time $t = 0$.

The velocity of the particle is described by

$$v = \sqrt{2x}\, e^{-x^2} \quad x \geq 0.1$$

where x is the displacement of the particle from the origin.

(i) Show that the acceleration of the particle is given by 2

$$a = e^{-2x^2}(1\text{-}4x^2)$$

(ii) Hence find the fastest speed attained by the particle. 2

(iii) Show that T, the time taken to travel from $x = 1$ to $x = 2$ can be expressed as 2

$$T = \int_1^2 \frac{1}{\sqrt{2x}} e^{x^2} dx$$

(iv) Use the trapezoidal rule with three function values to give an approximate value of T correct to the nearest whole number. 2

HSC Extension 2 Mathematics

402

Mechanics Solution

(i) $v^2 = 8 - 2x - x^2$

Let $v = 0$

$0 = x^2 + 2x - 8$ (1)

$0 = (x-2)(x+4)$

$x = 2$ and -4 (2)

oscillates between these.

(ii) $\frac{1}{2}v^2 = 4 - x - \frac{x^2}{2}$

$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = -1 - \frac{2x}{2}$

$a = -(x+1)$ (1)

(iii) Period $= \frac{2\pi}{n}$

$n = 1$ from above

$\therefore$ Period is 2π (1)

(iv) Travels 12m in one period $\therefore$ 5 periods

$= 5 \times 2\pi = 10\pi$ seconds (1)

HSC Extension 2 Mathematics

401

Mechanics Solution

$v^2 = n^2(a^2 - x^2)$

$3^2 = n^2(2^2 - 1^2)$

$9 = 3n^2$

$n^2 = 3 \quad n = \sqrt{3}$

$T = \frac{2\pi}{\sqrt{3}} = \frac{2\sqrt{3}\pi}{3}$

B

HSC Extension 2 Mathematics

404

Mechanics Solution

(i) $v = \sqrt{2x}\, e^{-x^2}, \quad x \geq 0.1$

$\frac{dv}{dx} = \sqrt{2x} \times -2x \times e^{-x^2} + e^{-x^2} \times \sqrt{2} \times \frac{1}{2} x^{-\frac{1}{2}}$

$= -2\sqrt{2}\, x^{3/2} e^{-x^2} + \frac{\sqrt{2}\, e^{-x^2}}{2\sqrt{x}}$ (1)

$a = v\frac{dv}{dx} = \sqrt{2x}\, e^{-x^2}\left[-2\sqrt{2}\, x^{3/2} e^{-x^2} + \frac{e^{-x^2}}{\sqrt{2x}}\right]$

$= -4x^2 e^{-2x^2} + e^{-2x^2}$ (2)

$= e^{-2x^2}(1 - 4x^2)$ as req'd

(ii) When $x = 0.1$, $a = v\frac{dv}{dx} > 0$. Particle increases speed until $a = 0$ ie: when $1 - 4x^2 = 0$

$4x^2 = 1$

$x^2 = \frac{1}{4}$

$x = 0.5$ (since $x \geq 0.1$) (1)

Sub. this into v gives

$v = \sqrt{2 \times 0.5} \times e^{-0.5^2}$

$v = e^{-0.25}$ is max. speed. (2)

(iii) $v = \frac{dx}{dt} = \sqrt{2x}\, e^{-x^2}$

$\therefore \int_1^2 \frac{e^{x^2}\, dx}{\sqrt{2x}} = \int dt$ (1)

$T = \int_1^2 \frac{e^{x^2}\, dx}{\sqrt{2x}}$ (2)

(iv)

x	1	1.5	2
$f(x)$	$\frac{e}{\sqrt{2}}$	$\frac{e^{2.25}}{\sqrt{3}}$	$\frac{e^4}{2}$

$\therefore T = \frac{h}{2}\left[y_0 + 2y_1 + y_2\right]$

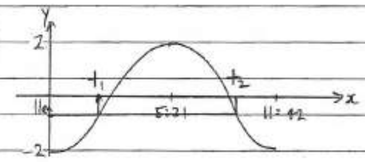
$= \frac{0.5}{2}\left[\frac{e}{\sqrt{2}} + \frac{2e^{2.25}}{\sqrt{3}} + \frac{e^4}{2}\right] = 10.04$

HSC Extension 2 Mathematics

403

Mechanics Solution

e)



(i) Period $= 12$ hrs 42 min

$= 762$ mins. (1)

(ii) $y = -2\cos nt$ (1)

$T = 762 = \frac{2\pi}{n}$

$\therefore n = \frac{\pi}{381}$

$\therefore y = -2\cos\frac{\pi t}{381}$ (2)

(iii) 1m above low tide $\Rightarrow y = -1$ on our graph

$-1 = -2\cos\frac{\pi t}{381}$ (1)

$\frac{1}{2} = \cos\frac{\pi t}{381}$

$\therefore \frac{\pi t}{381} = \frac{\pi}{3}, \frac{5\pi}{3}$

$t = \frac{381}{3}, \frac{5 \times 381}{3}$

$= 127, 635$

$= 2$ hrs 7 min and 10 hrs 35 min after 11am ie: safe between 1:07pm and 9:35pm (2)

HSC Extension 2 Mathematics — 405

Mechanics

A particle which was initially at rest, subsequently moves in a straight line along the x-axis. Its acceleration is given by $\ddot{x} = (2v+1)\ \mathrm{m\,s^{-2}}$, where $v\ \mathrm{m\,s^{-1}}$ is the velocity of the particle after t seconds. [2]

By expressing $\ddot{x}$ as $\dfrac{dv}{dt}$, find an expression for $v(t)$.

HSC Extension 2 Mathematics — 406

Mechanics

A flight controller is monitoring the progress of three helicopters as they move along straight flight paths through Sydney airspace. Let the origin O be the control tower of Kingsford-Smith Airport. All distances are measured in kilometres with the Cartesian axes Ox, Oy and Oz oriented due east, due north and vertically upwards respectively.

The positions of the three helicopters are represented by the points $P(3t-2, 5-t, \frac{3}{2}+\frac{t}{4})$, $Q(t-1, 2t-7, 1+\frac{t}{5})$ and $R(8-2t, t+1, 3-\frac{t}{2})$ respectively, where t represents the time in minutes after midday.

(i) Describe the location of the first helicopter relative to the airport control tower at midday. [1]

(ii) Find the direction vectors of the flight paths of each of the three helicopters. [1]

(iii) Helicopters P and Q pass through a common point A in space, but at different times. Find the coordinates of this point and the time each pilot reaches this location. [3]

(iv) Show that the third helicopter R is also on course to pass through point A and identify which of the other two helicopters it will collide with, unless the pilots are instructed to alter their courses. [1]

HSC Extension 2 Mathematics — 407

Mechanics

An object moves so that its displacement x metres at time t seconds is given by:

$$x = \cos 3t + 2\sin 3t.$$

(i) Show that the motion is simple harmonic by showing that it satisfies the differential equation $\ddot{x} = -n^2 x$, for some $n > 0$. [1]

(ii) Express x in the form $R\sin(3t+\alpha)$, where $R > 0$ and $0 \leq \alpha < \frac{\pi}{2}$. [2]

(iii) Hence find at what time, to the nearest hundredth of a second, the object first passes through $x = 2$. [1]

HSC Extension 2 Mathematics — 408

Mechanics

The sea level in a harbour is assumed to rise and fall with simple harmonic motion. On a certain day low tide occurs at 1500 hours when the depth of the water will fall to 8 metres. The next high tide will occur at 0330 hours with a depth of 18 metres. [4]

A ship wishes to enter the harbour that evening and needs a minimum depth of 12 metres of water to do so safely. Calculate, to the nearest minute, the earliest time it can enter the harbour that evening and the time by which it must leave the following morning.

HSC Extension 2 Mathematics

406

Mechanics Solution

(i) At midday $t=0$

$P\left(-2, 5, \frac{3}{2}\right)$

First helicopter is 2km west, 5km north and at an altitude of $1\frac{1}{2}$km. ✓

(ii) Direction Vector of first helicopter P is $\begin{bmatrix} 3 \\ -1 \\ \frac{1}{4} \end{bmatrix}$

Similarly for Q, direction vector is $\begin{bmatrix} 1 \\ 2 \\ \frac{1}{5} \end{bmatrix}$

for R direction vector is $\begin{bmatrix} -2 \\ 1 \\ -\frac{1}{2} \end{bmatrix}$ ✓ (Allow 1 numerical/sign error but 2 correct vectors for mark)

(iii) If helicopter P & Q pass through same point but at different times replace parameter t with λ and μ respectively

Find where paths of helicopters P & Q cross

$$\begin{bmatrix} 3\lambda-2 \\ 5-\lambda \\ \frac{3}{2}+\frac{\lambda}{4} \end{bmatrix} = \begin{bmatrix} \mu-1 \\ 2\mu-7 \\ 1+\frac{\mu}{5} \end{bmatrix}$$

$3\lambda-2 = \mu-1$ ①

$5-\lambda = 2\mu-7$ ② ✓

2×① − ②: $7\lambda-9=5$

$\lambda = 2$

sub into ①: $4 = \mu-1$

$\mu = 5$ ✓

check in ②: LHS $5-\lambda=3$

RHS $2\mu-7=3$

Should confirm consistent with z co-ordinate as well to ensure lines not SKEW

$\frac{3}{2}+\frac{\lambda}{4} = 1+\frac{\mu}{5}$ ③

LHS $\frac{3}{2}+\frac{2}{4}=2$

RHS $1+\frac{5}{5}=2$

hence paths of P & Q cross at A (4,3,2) with P passing through at 12:02pm and Q at 12:05pm ✓ (Must confirm consistency with 3rd co-ord not used to solve for λ & μ to gain full marks)

Confirm helicopter R passes through A

$$\begin{bmatrix} 8-2t \\ t+1 \\ 3-\frac{t}{2} \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 2 \end{bmatrix}$$

$8-2t=4$ ①, $2t=4$, $t=2$

$t+1=3$ ②, $t=2$

$3-\frac{t}{2}=2$ ③, $1=\frac{t}{2}$, $2=t$

hence helicopter R passes through A at 12:02pm and will crash with P unless one changes course

✓ Must show $t=2$ for all 3 eqns. Else substitute $t=2$ into R and confirm R(4,3,2) for the mark since "SHOW that"

HSC Extension 2 Mathematics

405

Mechanics Solution

$$\frac{dv}{dt} = 2v+1$$

$$\frac{dt}{dv} = \frac{1}{2v+1}$$

$$\left(\int dv\right) \quad t = \frac{1}{2}\ln|2v+1| + C \quad ✓$$

$t=0$ $v=0$ $\Rightarrow C=0$

$$t = \frac{1}{2}\ln|2v+1|$$

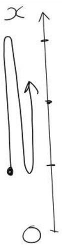
$$e^{2t} = 2v+1$$

$$v = \frac{1}{2}\left(e^{2t}-1\right) \quad ✓$$

HSC Extension 2 Mathematics

408

Mechanics Solution



$x=18$m $t=12\frac{1}{2}$hrs (3:30hrs)

$x=13$m (Centre)

$x=8$m, $t=0$ (15:00hrs)

Define x metres as depth

Let $t=0$hrs be 15:00

Hence amplitude of motion is 5m

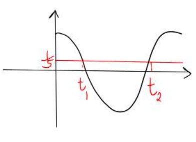
Time Period is 25 hrs

$\frac{2\pi}{n}=25 \Rightarrow n=\frac{2\pi}{25}$ ✓

Model depth of water using $x = 13-5\cos\frac{2\pi t}{25}$ ✓

Minimum depth of 12m $x \geq 12$ $\Rightarrow$ $5\cos\frac{2\pi t}{25} \leq 1$

$\cos\frac{2\pi t}{25} \leq \frac{1}{5}$



requires $t_1 \leq t \leq t_2$

at 12m, $\frac{2\pi t}{25} = \cos^{-1}\left(\frac{1}{5}\right)$

$\doteqdot 1.369\ldots, 2\pi-1.369$

hence

$1.369\ldots \leq \frac{2\pi t}{25} \leq 2\pi - 1.369\ldots$ ✓

$5.448\ldots \leq t \leq 25-5.448\ldots$

$5.448\ldots \leq t \leq 19.551\ldots$

5hrs 27mins $\leq t \leq$ 19hrs 33mins

Ship can safely enter harbour at 20:27 and must leave by 10:33 the following morning ✓

HSC Extension 2 Mathematics

407

Mechanics Solution

(i) $x = \cos 3t + 2\sin 3t$

$\dot{x} = -3\sin 3t + 6\cos 3t$

$\ddot{x} = -9\cos 3t - 18\sin 3t$

$= -9(\cos 3t + 2\sin 3t)$

$= -9x$ ✓ SHOW

hence SHM with $n=3$

(ii) Let $R\sin(3t+\alpha) = \cos 3t + 2\sin 3t$

$R\sin 3t\cos\alpha + R\cos 3t\sin\alpha = \cos 3t + 2\sin 3t$

Equate coeffs of $\sin 3t$: $R\cos\alpha = 2$ ①

$\cos 3t$: $R\sin\alpha = 1$ ② ✓

①^2 + ②^2: $R^2(\cos^2\alpha + \sin^2\alpha) = 4+1$

$R=\sqrt{5}$ $R>0$

hence $\sin\alpha = \frac{1}{\sqrt{5}}$ $\cos\alpha = \frac{2}{\sqrt{5}}$ $\alpha = \sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$ or $\cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$ or $\tan^{-1}\left(\frac{1}{2}\right)$

$x = \sqrt{5}\sin\left(3t + \sin^{-1}\frac{1}{\sqrt{5}}\right)$ ✓

(iii) $x=2$ $\Rightarrow$ $\sin(3t+\alpha) = \frac{2}{\sqrt{5}}$

$t>0$

$3t+\alpha > \sin^{-1}\frac{1}{\sqrt{5}}$

$\doteqdot 0.46\ldots$

$3t+\alpha = 1.107\ldots$

$3t = 0.6435\ldots$

$t = 0.2145\ldots$

$\doteqdot 0.21$ seconds ✓

HSC Extension 2 Mathematics

409

Mechanics

A particle of mass m is projected vertically upwards with initial speed u metres per second and experiences a resistance proportional to its speed v. That is, a resistive force of mkv Newtons acts in an opposite direction to its motion, where k is a positive constant. After T seconds the particle attains its maximum height H metres.

Let the acceleration due to gravity be g m/s^2. Define the origin as the point of projection and upwards as the positive direction for the motion. The acceleration of the particle can thus be given by the differential equation

$$\ddot{y} = -(g + kv),$$

where y is the height of the particle t seconds after the launch.

(i) Prove that $T = \dfrac{1}{k}\ln\left(\dfrac{g+ku}{g}\right)$. 2

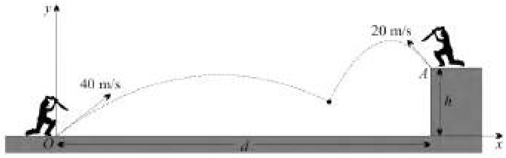
(ii) Show that $H = \dfrac{u - gT}{k}$. 3

(iii) Before considering the downward journey we will define a new origin as the point reached when the particle is at its maximum height. Also define downwards as the new positive direction for the motion. Using this revised coordinate system, write down the new differential equation for the acceleration of the particle during its downward journey. 1

HSC Extension 2 Mathematics

410

Mechanics



Two cricketers are trying to film a 'trick' by simultaneously each hitting a cricket ball and making the balls collide in mid-air. The first cricketer is at point O, at ground level, while the second is at the top of a nearby building at point A. The point A is a horizontal distance d metres away from point O and at a height h metres above the horizontal ground, as shown in the diagram above. 4

The ball hit from O has an initial speed of 40 m/s and is given an angle of projection of $\tan^{-1}\left(\frac{3}{4}\right)$. Define the origin at O with right and upwards as the positive directions for x and y respectively, as shown in the diagram above. It then follows that the six equations describing the motion of the ball hit by the first cricketer, t seconds after release are:

$$\ddot{x} = 0 \qquad \dot{x} = 32 \qquad x = 32t$$

$$\ddot{y} = -g \qquad \dot{y} = 24 - gt \qquad y = 24t - \frac{1}{2}gt^2$$

The other cricketer hits the second ball at the same instant, from point A towards O, with an initial speed of 20 m/s and an angle of projection of $\tan^{-1}\left(\frac{4}{3}\right)$. Assume that the motion of the two balls takes place in the same vertical plane, they behave as particles, experience no air resistance and that the cricketers are successful in making the two balls collide during their flights.

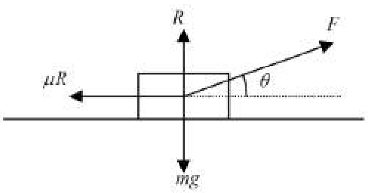
Find the comparable set of six equations for the motion of the second cricket ball and hence find the ratio of the two lengths d and h.

HSC Extension 2 Mathematics

411

Mechanics

An object with mass m is being pulled along horizontal ground with acceleration a by a constant force F acting at angle θ to the horizontal, as shown in the diagram. A force due to gravity of mg acts downwards and a normal force R acts upwards and a resistive force μR acts parallel to the ground to oppose the motion, where μ, the coefficient of friction, is a constant. 3



By balancing horizontal and vertical components of forces, show that $\mu = \dfrac{F\cos\theta - ma}{mg - F\sin\theta}$.

HSC Extension 2 Mathematics

412

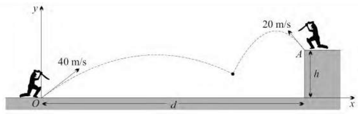
Mechanics

An object of mass 1 kg is projected vertically upwards from the origin with initial velocity V_0, so that the only forces acting on it are a force due to gravity and air resistance, which is has size $\dfrac{\dot{y}}{10}$. Taking y as the displacement upwards from the origin, and acceleration due to gravity to be 10 ms^{-2}, its equation of motion is: 3

$$\ddot{y} = -\left(10 + \frac{\dot{y}}{10}\right)$$

Show that the maximum height the object reaches, y_{max} is

$$y_{max} = 10\left(V_0 - 100\ln\left(\frac{100 + V_0}{100}\right)\right).$$



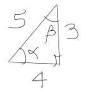
$\ddot{x}=0 \quad \dot{x}=32 \quad x=32t$

$\ddot{y}=-g \quad \dot{y}=24-gt \quad y=24t-\frac{1}{2}gt^2$

Taking origin at O and $t=0$ at time of projection

Define angles of projection α & β at O and A respectively

$\alpha=\tan^{-1}\left(\frac{3}{4}\right) \quad \beta=\tan^{-1}\left(\frac{4}{3}\right)$



For ball from A

$\ddot{x}_A=0$

$\dot{x}_A=-20\cos\beta = -20\times\frac{3}{5} = -12$

$x_A = d-12t$

$\ddot{y}_A=-g$

$\dot{y}_A = 20\sin\beta - gt = 20\times\frac{4}{5}-gt = 16-gt$

$y_A = 16t-\frac{1}{2}gt^2+h$

Balls collide at time when x displacements equal

$32t = d-12t$

$44t = d$

$t = \frac{d}{44}$ (1)

At this time y displacements also equal

$24t-\frac{1}{2}gt^2 = 16t-\frac{1}{2}gt^2+h$

$8t = h$

$t = \frac{h}{8}$ (2)

hence $\frac{d}{44}=\frac{h}{8}$

$\frac{d}{h}=\frac{44}{8}$

$\frac{d}{h}=\frac{11}{2}$

ratio of lengths $d:h = 11:2$

(i) $\frac{dv}{dt} = -(g+kv)$

$\int_u^0 \frac{dv}{g+kv} = -\int_0^T dt$

$\left[\frac{1}{k}\ln|g+kv|\right]_u^0 = [-t]_0^T$

$-T = \frac{1}{k}\ln g - \frac{1}{k}\ln(g+ku)$ SHOW

$T = \frac{1}{k}\left[\ln(g+ku)-\ln g\right]$

$T = \frac{1}{k}\ln\left(\frac{g+ku}{g}\right)$ as required

(ii) $\frac{v\,dv}{dy} = -(g+kv)$

$\int_u^0 \frac{v\,dv}{g+kv} = -\int_0^H dy$

$[-y]_0^H = \frac{1}{k}\int_u^0 \frac{kv+g-g}{g+kv}\,dv$

$-H = \frac{1}{k}\int_u^0 1-\frac{g}{g+kv}\,dv$

$H = -\frac{1}{k}\left[v-\frac{g}{k}\ln(g+kv)\right]_u^0$

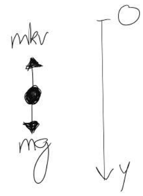
$H = -\frac{1}{k}\left[\left(-\frac{g}{k}\ln g\right)-\left(u-\frac{g}{k}\ln(g+ku)\right)\right]$

$H = -\frac{1}{k}\left[\frac{g}{k}\ln\left(\frac{g+ku}{g}\right)-u\right]$ SHOW

$= -\frac{1}{k}\left[gT-u\right]$

$H = \frac{1}{k}(u-gT)$ as required

(iii)



Eqn of motion ↓ $mg - mkv = m\ddot{y}$

$\ddot{y} = g-kv$

$\ddot{y} = -\left(10+\frac{\dot{y}}{10}\right)$

$\dot{y}\frac{d\dot{y}}{dy} = -\left(10+\frac{\dot{y}}{10}\right)$

$-\int_{v_0}^{0} \frac{\dot{y}}{10+\frac{\dot{y}}{10}}\,d\dot{y} = \int_0^{y_{max}} dy$

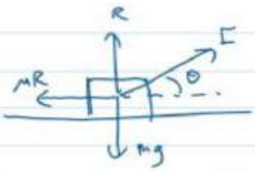
$y_{max}-0 = 10\int_0^{v_0}\frac{\dot{y}}{100+\dot{y}}\,d\dot{y}$

$y_{max} = 10\int_0^{v_0} 1-\frac{100}{100+\dot{y}}\,d\dot{y}$

$= 10\left[\dot{y}\right]_0^{v_0} - 1000\left[\ln|100+\dot{y}|\right]_0^{v_0}$

$= 10v_0 - 1000\ln\left(\frac{100+v_0}{100}\right)$

$= 10\left(v_0-100\ln\left(\frac{100+v_0}{100}\right)\right)$



Balancing horizontal forces:

(1) $F\cos\theta - \mu R = ma \rightarrow \mu R = F\cos\theta - ma$

Balancing forces vertically:

(2) $R + F\sin\theta - mg = 0$

From (1), $\mu = \frac{F\cos\theta - ma}{R}$

From (2), $R = mg - F\sin\theta$

$\therefore \mu = \frac{F\cos\theta - ma}{mg - F\sin\theta}$

HSC Extension 2 Mathematics

413

Mechanics

★★★

A motorbike that is initially stationary at the origin has mass 400 kg and begins to travel horizontally with a constant propulsive force of 800N. A resistive force opposes the object's motion, proportional to the square of the velocity, such that its equation of motion is:

$$400\ddot{x} = 800 - kv^2$$

Where k is a constant, v is its velocity, and x is displacement in the direction of the motion after t seconds.

(i) Show that $\dfrac{1}{a^2-b^2} = \dfrac{1}{2a}\left(\dfrac{1}{a+b} + \dfrac{1}{a-b}\right)$. **1**

(ii) Given that the terminal velocity of the motorbike is 40 ms^{-1}, show that the object's velocity is given by **3**

$$v = 40\left(\frac{e^{0.1t}-1}{e^{0.1t}+1}\right).$$

(iii) Find an expression for displacement x in terms of t. **3**

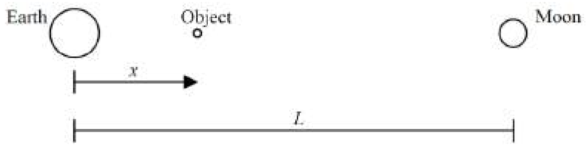
HSC Extension 2 Mathematics

414

Mechanics

★★★

Consider an object travelling from the Earth to the moon. The distance between the centres of the Earth and moon is L and can be assumed to remain constant. The distance of the object from the centre of the Earth in the direction of the moon is x, where $0 < x < L$. Initially the object is at $x = 0.5L$ and is travelling with velocity u ms^{-1}, where $u > 0$.



The equation of motion of the object is given by:

$$\ddot{x} = -\frac{81k}{x^2} + \frac{k}{(L-x)^2}, \text{ where } k \text{ is a constant.}$$

(i) Determine that the value of x for which the object has zero acceleration is $x = 0.9L$. **2**

(ii) Show that $v^2 = \dfrac{162k}{x} + \dfrac{2k}{L-x} + u^2 - \dfrac{328k}{L}$. **2**

(iii) By substituting $x = \mu L$ and given that $\dfrac{u^2 L}{k} = 80$, deduce that the object is stationary at $\mu \approx 0.67$. **3**

(iv) Briefly describe the object's motion for $t > 0$. **1**

HSC Extension 2 Mathematics

415

Mechanics

★★★

A body of unit mass falls under gravity through a resisting medium. The body falls from rest from a height above the ground. The resistance to its motion is $\dfrac{1}{100}v^2$ where v m/s is the speed of the body when it has fallen a distance x m.
The acceleration due to gravity is g m/s^2.

(i) Show that the equation of motion of the body is $\ddot{x} = g - \dfrac{1}{100}v^2$ **1**

(ii) Show that the terminal velocity V m/s of the body is given by $V = 10\sqrt{g}$ **1**

(iii) Hence show that $v^2 = V^2\left(1 - e^{-\frac{x}{50}}\right)$ **3**

(iv) Find the distance fallen in metres until the body reaches a velocity equal to half of its terminal velocity.
(You may assume this occurs before the body reaches the ground.) **2**

HSC Extension 2 Mathematics

416

Mechanics

★★

A body of mass m kg is travelling in a horizontal straight line so that the resultant force on the body is a resistance force of magnitude $\dfrac{1}{10}m\sqrt{1+v}$ when its speed is v m/s. Initially, the speed of the body is 15 m/s.

(i) Find the time taken for the body to come to rest. **2**

(ii) Find the distance travelled by the body in coming to rest. **3**

HSC Extension 2 Mathematics

414

Mechanics Solution

(i) Solve $\ddot{x} = 0$:

$$0 = \frac{-81k}{x^2} + \frac{k}{(L-x)^2}$$

$$\frac{81}{x^2} = \frac{1}{(L-x)^2}$$

$$\frac{x^2}{81} = (L-x)^2$$

$$\frac{x}{9} = \pm(L-x)$$

$x = 9L - 9x$ | $x = 9x - 9L$
$10x = 9L$ | $8x = 9L$
$x = \frac{9}{10}L$ | $x = \frac{9}{8}L$
 | but $0 < x < L$

$\therefore x = \frac{9}{10}L = 0.9L$

(ii) $\frac{d}{dx}\left(\frac{1}{2}v^2\right) = -81kx^{-2} + k(L-x)^{-2}$

$$\frac{1}{2}v^2 = \frac{81k}{x} + \frac{k}{L-x} + \frac{c}{2}$$

$$v^2 = \frac{162k}{x} + \frac{2k}{L-x} + c$$

@ $t = 0$, $x = \frac{1}{2}L$ $v = u$:

$$u^2 = \frac{324k}{L} + \frac{4k}{L} + c$$

$$c = u^2 - \frac{328k}{L}$$

$$\therefore v^2 = \frac{162k}{x} + \frac{2k}{L-x} + u^2 - \frac{328k}{L}$$

(iii) @ $v = 0$:

$$0 = \frac{162k}{\mu L} + \frac{2k}{L(1-\mu)} + u^2 - \frac{328k}{L}$$

$$0 = 162(1-\mu) + 2\mu + \frac{u^2L}{k}\mu(1-\mu) - 328\mu(1-\mu)$$

$$= 162 - 162\mu + 2\mu + (80-328)\mu(1-\mu)$$

$$0 = 162 - 160\mu - 248\mu + 248\mu^2$$

$$248\mu^2 - 408\mu + 162 = 0$$

$$\mu = \frac{408 \pm \sqrt{408^2 - 4(248)(162)}}{2 \times 248}$$

$$\mu \approx 0.98,\ 0.67$$

i.e. $v = 0$ @ $\mu \doteq 0.67$

(iv) The object slows down until it comes to a stop @ $x \approx 0.67L$, it then changes direction and accelerates towards Earth.

HSC Extension 2 Mathematics

413

Mechanics Solution

$$400\ddot{x} = 800 - kv^2$$

(i) $\frac{1}{2a}\left(\frac{1}{a+b} + \frac{1}{a-b}\right) = \frac{1}{2a}\left(\frac{a-b+a+b}{(a+b)(a-b)}\right) = \frac{1}{2a}\left(\frac{2a}{a^2-b^2}\right) = \frac{1}{a^2-b^2}$

(ii) $v_T = 40$

@ $\ddot{x} = 0$,

$$0 = 800 - k(40)^2$$

$$k = \frac{800}{40^2}$$

$$k = \frac{1}{2}$$

$$400\frac{dv}{dt} = 800 - \frac{1}{2}v^2$$

$$\int_0^v \frac{400}{800 - \frac{1}{2}v^2}\,dv = \int_0^t dt$$

$$800\int_0^v \frac{1}{1600 - v^2}\,dv = t - 0$$

$$\frac{800}{2(40)}\int_0^v \frac{1}{40+v} + \frac{1}{40-v}\,dv = t$$

$$10\left[\ln\left(\frac{40+v}{40-v}\right)\right]_0^v = t$$

$$\ln\left(\frac{40+v}{40-v}\right) = 0.1t$$

$$40 + v = e^{0.1t}(40 - v)$$

$$v(e^{0.1t} + 1) = 40(e^{0.1t} - 1)$$

$$v = 40\left(\frac{e^{0.1t} - 1}{e^{0.1t} + 1}\right)$$

(iii) $\frac{dx}{dt} = 40\left(\frac{e^{0.1t} - 1}{e^{0.1t} + 1}\right)$

$$\int_0^x dx = 40\int_0^t \frac{e^{0.1t} - 1}{e^{0.1t} + 1}\,dt$$

$$x - 0 = 40\int_0^t 1 - \frac{2}{e^{0.1t} + 1}\,dt$$

$$x = 40\int_0^t 1 + \left(\frac{2}{0.1}\right)\frac{(-0.1)e^{-0.1t}}{1 + e^{-0.1t}}\,dt$$

$$= 40\left[t + 20\ln(1 + e^{-0.1t})\right]_0^t$$

$$= 40t + 800\ln\left(\frac{1 + e^{-0.1t}}{2}\right)$$

HSC Extension 2 Mathematics

416

Mechanics Solution

(i) $F = ma$

$$-\frac{1}{10}\cancel{m}\sqrt{1+v} = \cancel{m}\ddot{x}$$

$$\ddot{x} = \frac{dv}{dt} = -\frac{1}{10}\sqrt{1+v}$$

$$\frac{dt}{dv} = \frac{-10}{\sqrt{1+v}} \checkmark$$

$$\int_0^T dt = \int_{15}^0 -10(1+v)^{-1/2}\,dv$$

$$[t]_0^T = 10\left[\frac{(1+v)^{1/2}}{1/2}\right]_0^{15}$$

$$T - 0 = 20(\sqrt{16} - \sqrt{1})$$

$$\therefore T = 60 \text{ seconds} \checkmark$$

(ii) $\ddot{x} = v\frac{dv}{dx} = -\frac{1}{10}\sqrt{1+v}$

$$\frac{dv}{dx} = \frac{-\sqrt{1+v}}{10v}$$

$$\frac{dx}{dv} = \frac{-10v}{\sqrt{1+v}} \checkmark$$

$$\int_0^D dx = \int_{15}^0 \frac{-10v}{\sqrt{1+v}}\,dv$$

$$[x]_0^D = 10\int_0^{15} \frac{1+v-1}{\sqrt{1+v}}\,dv$$

$$D - 0 = 10\int_0^{15} (1+v)^{1/2} - (1+v)^{-1/2}\,dv \checkmark$$

$$D = \left[\frac{(1+v)^{3/2}}{3/2} - \frac{(1+v)^{1/2}}{1/2}\right]_0^{15}$$

$$= \frac{2}{3}(16\sqrt{16} - 1\sqrt{1}) - 2(\sqrt{16} - \sqrt{1})$$

$$\therefore D = 360 \text{ m} \checkmark$$

HSC Extension 2 Mathematics

415

Mechanics Solution

(i) $F = ma$ $\quad \downarrow g \quad \uparrow \frac{1}{100}v^2$

$$g - \frac{1}{100}v^2 = 1 \times \ddot{x}$$

$$\therefore \ddot{x} = g - \frac{1}{100}v^2$$

(ii) When $\ddot{x} = 0$, $v = V > 0$

$$0 = g - \frac{1}{100}V^2$$

$$\frac{1}{100}V^2 = g$$

$$V^2 = 100g$$

$$\therefore V = 10\sqrt{g}$$

(iii) $\ddot{x} = v\frac{dv}{dx} = g - \frac{1}{100}v^2$

$$\frac{dv}{dx} = \frac{100g - v^2}{100v}$$

$$\frac{dx}{dv} = \frac{100v}{100g - v^2}$$

$$x = \int \frac{100v}{100g - v^2}\,dv$$

$$= -50\ln|100g - v^2| + c$$

At $t = 0$, $x = 0$, $v = 0$

$$0 = -50\ln|100g - 0^2| + c$$

$$c = 50\ln 100g$$

$$\therefore x = 50\ln\left|\frac{100g}{100g - v^2}\right|$$

$$\frac{x}{50} = \ln\left|\frac{100g}{100g - v^2}\right|$$

$$e^{\frac{x}{50}} = \frac{100g}{100g - v^2}$$

$$100g - v^2 = 100g\,e^{-x/50}$$

$$v^2 = 100g(1 - e^{-x/50})$$

$$\therefore v^2 = V^2(1 - e^{-x/50})$$

(iv) Sub $v = V/2$

$$\frac{\cancel{V^2}}{4} = \cancel{V^2}(1 - e^{-x/50})$$

$$e^{-x/50} = 3/4$$

$$-\frac{x}{50} = \ln\frac{3}{4}$$

$$x = 50\ln 4/3$$

$$\therefore x = 14.38 \text{ m (2dp)}$$

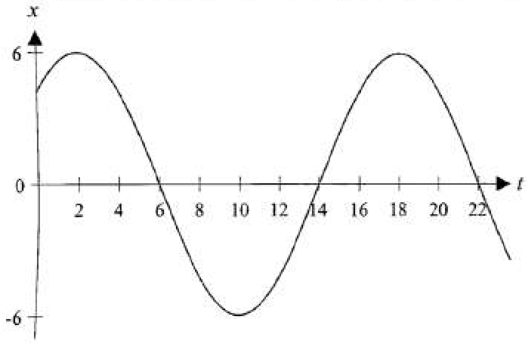
HSC Extension 2 Mathematics

417

Mechanics

★★★

The graph below shows the displacement x cm from the centre of motion at time t seconds for a particle performing simple harmonic motion in a straight line.



(i) Express displacement as a function of time in the form $x = A\sin(nt + \alpha)$ **2**

(ii) Find the distance travelled by the particle in the first 30 seconds of its motion after observation began at time $t = 0$. **2**

HSC Extension 2 Mathematics

418

Mechanics

★★

When an object moves through water, it experiences resistance proportional to its speed.

(i) If an object of unit mass moving at 10 m/s in water experiences a resistance of 40 Newtons,
write an expression for the acceleration due to resistance, in terms of v. **1**

(ii) A metal ball of unit mass is released from rest in water and immediately sinks. Given that $g = 10$ m/s^2, and after t seconds the velocity of the ball is v m/s, find the velocity of the metal ball as a function of t. **3**

HSC Extension 2 Mathematics

419

Mechanics

★★★

A particle moves in a straight line. At time t seconds, its displacement from a fixed origin is x metres and its velocity is v m/s. The acceleration of the particle is given by $\ddot{x} = x + 3$.
At $t = 0$, the particle is at the origin and moving with velocity 3 m/s.

(i) Show that $v = x + 3$ **2**

(ii) Find an expression for x, the displacement of the particle, in terms of t. **2**

HSC Extension 2 Mathematics

420

Mechanics

★★★

A particle P of mass m kg is released from rest and falls under its own weight.

The particle is subject to air resistance of magnitude kmv^2 N, where v ms^{-1} is the speed of the particle after it has travelled x m and k is a positive constant.

(i) Show that $v^2 = \left(1 - e^{-2kx}\right)\frac{g}{k}$. **3**

(ii) Hence determine the terminal velocity, V_T, of the particle. **1**

(iii) Show that the particle attains a speed of $\frac{1}{2}V_T$ after it has fallen a distance of $\frac{1}{2k}\ln\left(\frac{4}{3}\right)$ metres. **2**

HSC Extension 2 Mathematics

418

Mechanics Solution

(i) Resistance $F = kv$

$$40 = k \times 10$$
$$k = 4$$

Resultant force $F = ma = -kv$

$$ma = -4v$$
$$a = -\frac{4}{m}v$$

(ii) $\frac{dv}{dt} = g - kv$ $\quad (m = 1)$

$$= 10 - 4v$$
$$\frac{dt}{dv} = \frac{1}{10-4v}$$
$$\int_0^t dt = \int_0^v \frac{1}{10-4v}\,dv$$
$$\left[t\right]_0^t = \left[-\frac{1}{4}\ln|10-4v|\right]_0^v$$
$$t = -\frac{1}{4}\left[\ln|10-4v| - \ln 10\right]$$
$$t = \frac{1}{4}\ln\left|\frac{10}{10-4v}\right|$$
$$e^{4t} = \frac{10}{10-4v}$$
$$10 - 4v = \frac{10}{e^{4t}}$$
$$-4v = \frac{10}{e^{4t}} - 10$$
$$v = \frac{5}{2} - \frac{5}{2e^{4t}}$$

HSC Extension 2 Mathematics

417

Mechanics Solution

(i) $x = A\sin(nt + \alpha)$

$A = 6$

$T = \frac{2\pi}{n} = 16$

$n = \frac{\pi}{8}$ ✓

At $t = 2$, $x = 6$

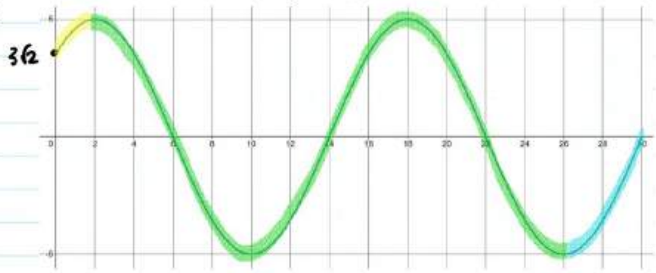
$6 = 6\sin\left(\frac{\pi}{4} + \alpha\right)$

$\frac{\pi}{4} + \alpha = \sin^{-1}(1) = \frac{\pi}{2}$

$\alpha = \frac{\pi}{4}$

$\therefore x = 6\sin\left(\frac{\pi t}{8} + \frac{\pi}{4}\right)$ ✓

(ii)



At $t = 0$, $x = 6\sin\left(0 + \frac{\pi}{4}\right) = \frac{6}{\sqrt{2}} = 3\sqrt{2}$ ✓

Distance $= (6 - 3\sqrt{2}) + 3 \times 12 + 6 = 48 - 3\sqrt{2}$ metres

HSC Extension 2 Mathematics

420

Mechanics Solution

(i) $ma = mg - kmv^2$ $\quad$ where $k > 0$

$$a = g - kv^2$$
$$v\frac{dv}{dx} = g - kv^2$$
$$\frac{dx}{dv} = \frac{v}{g - kv^2}$$
$$x = -\frac{1}{2k}\ln\left(g - kv^2\right) + c$$

$x = 0$ when $v = 0$ $\quad \therefore c = \frac{1}{2k}\ln g$

$$x = \frac{1}{2k}\ln g - \frac{1}{2k}\ln\left(g - kv^2\right)$$
$$x = \frac{1}{2k}\ln\left(\frac{g}{g - kv^2}\right)$$
$$\therefore e^{2kx} = \frac{g}{g - kv^2}$$
$$g - kv^2 = ge^{-2kx}$$
$$v^2 = \frac{g}{k}\left(1 - e^{-2kx}\right)$$

(ii) As $x \to \infty, e^{-2kx} \to 0$

$$\therefore v^2 \to \frac{g}{k} \qquad v_T = \sqrt{\frac{g}{k}}$$

(iii) Substituting for x.

$$v^2 = \left(1 - e^{-2k\left(\frac{1}{2k}\ln\frac{4}{3}\right)}\right)\frac{g}{k} = \left(1 - e^{\frac{3}{4}}\right)\frac{g}{k} = \left(1 - \frac{3}{4}\right)\frac{g}{k}$$
$$\therefore v = \frac{1}{2}\sqrt{\frac{g}{k}} = \frac{1}{2}v_T$$

HSC Extension 2 Mathematics

419

Mechanics Solution

(i)

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = x + 3$$
$$\frac{1}{2}v^2 = \int x + 3\,dx$$
$$= \frac{1}{2}x^2 + 3x + c \qquad x = 0 \text{ gives } v = 3$$
$$\frac{1}{2}\left(3^2\right) = c = \frac{9}{2}$$
$$\therefore \frac{1}{2}v^2 = \frac{x^2}{2} + 3x + \frac{9}{2}$$
$$v^2 = \pm(x+3)^2$$

Initial conditions give positive velocity towards the positive side of the x-axis, so v can only ever be positive. Hence $v = x + 3$

(ii)

$$\frac{dx}{dt} = x + 3 \to \frac{dt}{dx} = \frac{1}{x+3}$$
$$t = \ln(x+3) + c \qquad t = 0 \text{ gives } x = 0$$
$$0 = \ln 3 + c \quad \to \quad c = -\ln 3$$
$$\therefore t = \ln\left(\frac{x+3}{3}\right)$$
$$e^t = \frac{x+3}{3}$$
$$x = 3e^t - 3$$

HSC Extension 2 Mathematics

421

Mechanics

A particle is moving on a straight line. Its velocity v is given by $v^2 = 4\left(2x - x^2\right)$ where x is its displacement from a point O on the line.

(i) Show that its acceleration is given by $\ddot{x} = -4\left(x - 1\right)$. **2**

(ii) Explain why this particle moves in simple harmonic motion. **1**

(iii) Find the maximum speed of the particle. **1**

HSC Extension 2 Mathematics

422

Mechanics

Show that a particle which moves according to the equation $v^2 = 36 - 6x - 2x^2$ is undergoing simple harmonic motion, where v is the velocity of the particle and x is the displacement of the particle. **2**

HSC Extension 2 Mathematics

423

Mechanics

A particle of unit mass is projected vertically upwards from the ground at a speed of V ms^{-1}. The particle is acted on by both gravity, and air resistance of magnitude $\frac{v^2}{40}$, where v is the velocity of the particle measured in ms^{-1}. After t seconds, the particle's height from the ground, is x metres.

(i) Draw a force diagram illustrating all the forces acting on the particle while the particle is moving upwards and derive the equation of motion $\ddot{x} = -\left(g + \frac{v^2}{40}\right)$. **2**

(ii) Show that the greatest height the particle reaches is $h = 20\ln\left(\frac{40g + V^2}{40g}\right)$. **3**

(iii) Find the time taken to reach this greatest height. **3**

Having reached its maximum height, the particle falls back down towards its initial point of projection. Assume that only gravity and air resistance act on the particle.

(iv) Find the terminal velocity of the particle on its way down. **2**

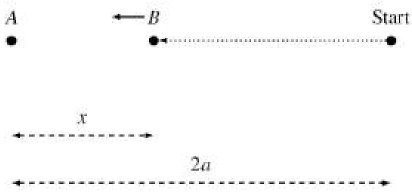
HSC Extension 2 Mathematics

424

Mechanics

Two particles with opposite charges are attracted to each other with a force numerically equal to $\frac{k^2}{x^2}$, where x is their distance apart in metres.

Particle A is fixed and particle B, of mass m kg, is released at a distance $2a$ metres from A, as shown in the diagram below.



(i) Show that the speed v of particle B can be given by $v^2 = \frac{2k^2}{m}\left(\frac{1}{x} - \frac{1}{2a}\right)$. **3**

(ii) Find the time taken for particle B to reach halfway to A from the start. **5**

HSC Extension 2 Mathematics

422

Mechanics Solution

$v^2 = 36 - 6x - 2x^2$

$a = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$

$= \frac{d}{dx}\left(18 - 3x - x^2\right)$

$= -3 - 2x$

$a = -2\left(x - \frac{3}{2}\right)$

which is of the form $a = -n^2(x - x_0)$

Hence, particle moves in SHM.

HSC Extension 2 Mathematics

421

Mechanics Solution

(i) $\frac{1}{2}v^2 = 2\left(2x - x^2\right)$

$\frac{d}{dx}\frac{1}{2}v^2 = 2(2 - 2x)$

$= -4(x - 1)$

(ii) $\ddot{x}$ is in the form $-n^2(x - c)$

Therefore motion is SHM.

(iii) $\ddot{x} = 0$ when $x = 1$

$\therefore v^2 = 4\left(2(1) - (1)^2\right) = 4$

$\therefore$ Max speed $= 2ms^{-1}$

HSC Extension 2 Mathematics

424

Mechanics Solution

i $m\ddot{x} = \frac{-k^2}{x^2}$ (Newton's 2nd law)

$\ddot{x} = \frac{-k^2}{mx^2}$

$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = -\frac{k^2}{mx^2}$

$\frac{1}{2}v^2 = \frac{-k^2}{m}\int \frac{1}{x^2}\,dx$

$v^2 = \frac{-2k^2}{m}\int \frac{1}{x^2}\,dx = \frac{2k^2}{mx} + C$

when $x = 2a$, $V = 0$

$\therefore 0 = \frac{2k^2}{m(2a)} + C \quad C = \frac{-k^2}{ma}$

$\therefore v^2 = \frac{2k^2}{mx} - \frac{k^2}{ma} = \frac{2k^2}{m}\left(\frac{1}{x} - \frac{1}{2a}\right)$

ii $v^2 = \frac{2k^2}{m}\left(\frac{1}{x} - \frac{1}{2a}\right) = \frac{2k^2}{m}\left(\frac{2a - x}{2ax}\right)$

$v = -\sqrt{\frac{2k^2}{m}} \times \sqrt{\frac{2a - x}{2ax}}$ (v is opposite to x)

$\frac{dx}{dt} = -\sqrt{\frac{2k^2}{m}} \times \sqrt{\frac{2a - x}{2ax}} = -\sqrt{\frac{k^2}{ma}} \times \sqrt{\frac{2a - x}{x}}$

$\int \frac{\sqrt{x}}{\sqrt{2a - x}}\,dx = -\sqrt{\frac{k^2}{ma}}\int dt$

$\int \frac{x}{\sqrt{2ax - x^2}}\,dx = \frac{-kt}{\sqrt{ma}} + C_1 \quad (k > 0)$

Consider LHS:

$\int \frac{x}{\sqrt{2ax - x^2}}\,dx$

$= \int \frac{-2x + 2a}{\sqrt{2ax - x^2}} \cdot \left(\frac{-1}{2}\right)dx + \int \frac{a}{\sqrt{2ax - x^2}}\,dx$

$= \frac{-1}{2}\int \frac{2a - 2x}{\sqrt{2ax - x^2}}\,dx + \int \frac{a}{\sqrt{a^2 - (x^2 - 2ax + a^2)}}\,dx$

$= \frac{-1}{2}\int \frac{2a - 2x}{\sqrt{2ax - x^2}}\,dx + \int \frac{a}{\sqrt{a^2 - (x - a)^2}}\,dx$

$= \frac{-1}{2} \cdot \left(2ax - x^2\right)^{\frac{1}{2}} \cdot 2 + a\sin^{-1}\left(\frac{x - a}{a}\right) + C_2$

$= a\sin^{-1}\left(\frac{x - a}{a}\right) - \sqrt{2ax - x^2} + C_1$

$a\sin^{-1}\left(\frac{x - a}{a}\right) - \sqrt{2ax - x^2} = \frac{-kt}{\sqrt{ma}} + C_3$

when $t = 0$, $x = 2a$ $\therefore \frac{a\pi}{2} = C_3$

$\therefore a\sin^{-1}\left(\frac{x - a}{a}\right) - \sqrt{2ax - x^2} = \frac{-kt}{\sqrt{ma}} + \frac{a\pi}{2}$

$\therefore$ when $x = a$, $-\sqrt{a^2} = \frac{-kt}{\sqrt{ma}} + \frac{a\pi}{2}$

$\therefore \frac{kt}{\sqrt{ma}} = \frac{a\pi}{2} + a$ (since $a > 0$)

$t = \frac{\sqrt{ma}\,a}{k}\left(\frac{\pi}{2} + 1\right)$

HSC Extension 2 Mathematics

423

Mechanics Solution

(i) $\uparrow +ve$

$\downarrow mg$

$\downarrow \frac{v^2}{40}$

OR "since $m = 1$"

$\downarrow g$

$\downarrow \frac{v^2}{40}$

$m\ddot{x} = -mg - \frac{v^2}{40}$ by Newton's 2nd law

since $m = 1$

$\ddot{x} = -g - \frac{v^2}{40}$

$= -\left(g + \frac{v^2}{40}\right)$

OR $\ddot{x} = -g - \frac{v^2}{40m}$

$m = 1$

$\therefore \ddot{x} = -g - \frac{v^2}{40}$

$= -\left(g + \frac{v^2}{40}\right)$

(ii) Greatest height reached when $\dot{x} = 0$

$v\frac{dv}{dx} = -\left(g + \frac{v^2}{40}\right)$

$v\frac{dv}{dx} = -\left(\frac{40g + v^2}{40}\right)$

$\int_V^0 \frac{40v}{40g + v^2}\,dv = -\int_0^h dx$

$20\left[\ln\left|40g + v^2\right|\right]_V^0 = \left[-x\right]_0^h$

$20\left(\ln(40g) - \ln\left(40g + V^2\right)\right) = -h + 0$, since $40g + v^2 > 0$

$h = 20\left(\ln\left(40g + V^2\right) - \ln(40g)\right)$

$= 20\ln\left(\frac{40g + V^2}{40g}\right)$

(iii) $\frac{dv}{dt} = -\left(\frac{40g + v^2}{40}\right)$

$\int_V^0 \frac{40}{40g + v^2}\,dv = -\int_0^T dt$ where T is time to reach greatest height

$\left[\frac{40}{\sqrt{40g}}\tan^{-1}\frac{v}{\sqrt{40g}}\right]_V^0 = \left[-t\right]_0^T$

$\sqrt{\frac{40}{g}}\tan^{-1}\frac{0}{\sqrt{40g}} - \sqrt{\frac{40}{g}}\tan^{-1}\frac{V}{\sqrt{40g}} = -T + 0$

$T = \sqrt{\frac{40}{g}}\tan^{-1}\frac{V}{\sqrt{40g}}$

(iv) $\uparrow \frac{v^2}{40}$

$\downarrow mg$

$\downarrow +ve$

$m\ddot{x} = mg - \frac{v^2}{40}$

$\ddot{x} = g - \frac{v^2}{40m}$, $m = 1$

$\therefore \ddot{x} = g - \frac{v^2}{40}$

Terminal velocity when $\ddot{x} = 0$

$\therefore \frac{v^2}{40} = g$

$v^2 = 40g$

$v = \sqrt{40g}$ since downwards motion positive

HSC Extension 2 Mathematics

425

Mechanics

A particle moves in a straight line with initial displacement $x = 0$. The velocity of the particle is given by $v = 2e^{-\frac{x}{2}}(x+1)^2$, where velocity is in metres per second.

(i) Show that the acceleration of the particle is given by $a = 2e^{-x}(x+1)^3(3-x)$. [2]

(ii) Hence find the displacement for which the maximum velocity of the particle will occur, justifying your answer. [2]

HSC Extension 2 Mathematics

426

Mechanics

A physicist drops an object with mass m kg from the top of a cliff. As the object falls, it experiences a force due to gravity of magnitude $10m$ Newtons and a resistive force of magnitude mkv^2 Newtons, where v is the velocity of the object in metres per second and k is a positive constant.

The vertical displacement of the object y metres from where it is dropped satisfies

$$m\ddot{y} = 10m - mkv^2 \,,$$

where the downwards direction is taken as positive.

(i) Let V be the terminal velocity of the object. Find an expression for V in terms of k, and hence show that $\dfrac{dv}{dt} = k(V^2 - v^2)$. [2]

(ii) Use part (b)(ii) to show that [3]

$$v = V \times \frac{e^{2Vkt} - 1}{e^{2Vkt} + 1} \,.$$

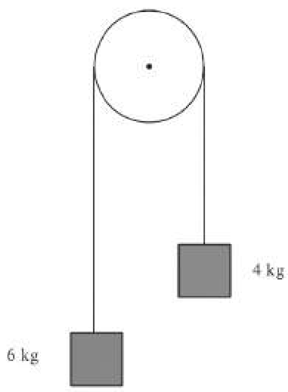
(iii) The physicist finds that it takes the object 1 second to reach half of its terminal velocity. Find the value of k. Give your answer correct to two decimal places. [2]

HSC Extension 2 Mathematics

427

Mechanics

A light inextensible string passes over a smooth pulley.
Attached to each end of the strings are masses of 4 kg and 6 kg, as shown.



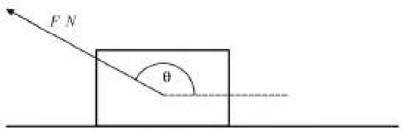
What is the acceleration of the larger mass?

HSC Extension 2 Mathematics

428

Mechanics

A block of mass $2\sqrt{2}$ kg, is pulled along a smooth horizontal plane by a force F as shown in th diagram below.



Which of the following set of conditions produces the largest magnitude of acceleration for the mass?

The information is presented as (F, θ).

A. $\left(\sqrt{2}, \pi\right)$

B. $\left(2\sqrt{3}, \dfrac{5\pi}{6}\right)$

C. $\left(2\sqrt{3}, \dfrac{3\pi}{4}\right)$

D. $\left(5, \dfrac{2\pi}{3}\right)$

HSC Extension 2 Mathematics

426

Mechanics Solution

(i) $m\ddot{y} = 10m - mkv^2$

$\ddot{y} = 10 - kv^2$

Terminal velocity when $\ddot{y} = 0$:

$kV^2 = 10$

$V = \sqrt{\frac{10}{k}}$, taking down as positive

$\ddot{y} = k\left(\frac{10}{k} - v^2\right)$

$\frac{dv}{dt} = k(V^2 - v^2)$

(ii) $\int \frac{dv}{V^2 - v^2} = \int k\,dt$

$\frac{1}{2V}\ln\left|\frac{V+v}{V-v}\right| = kt + c$ from (b)(ii)

$\ln\left|\frac{V+v}{V-v}\right| = 2Vkt + c_2$

$\left|\frac{V+v}{V-v}\right| = e^{2Vkt + c_2}$

$\frac{V+v}{V-v} = e^{c_2} \times e^{2Vkt}$ as $v < V$

When $t = 0$, $v = 0$: $\frac{V+0}{V-0} = e^{c_2} \times 1$

$e^{c_2} = 1$

$\therefore \frac{V+v}{V-v} = e^{2Vkt}$

$V + v = (V - v)e^{2Vkt}$

$v(1 + e^{2Vkt}) = V(e^{2Vkt} - 1)$

$v = V\frac{e^{2Vkt} - 1}{e^{2Vkt} + 1}$

(iii) When $t = 1$, $v = \frac{1}{2}V$:

$\frac{1}{2}\cancel{V} = \cancel{V}\frac{e^{2Vk} - 1}{e^{2Vk} + 1}$

$e^{2Vk} + 1 = 2e^{2Vk} - 2$

$e^{2Vk} = 3$

$2Vk = \ln 3$

but $V = \sqrt{\frac{10}{k}}$:

$\sqrt{\frac{10}{k}} \times k = \frac{1}{2}\ln 3$

$\sqrt{10} \times \sqrt{k} = \frac{1}{2}\ln 3$

$k = \left(\frac{\ln 3}{2\sqrt{10}}\right)^2 = 0.03$

HSC Extension 2 Mathematics

425

Mechanics Solution

(i) $v = 2e^{-\frac{x}{2}}(x+1)^2$

$v^2 = 4e^{-x}(x+1)^4$

$\frac{1}{2}v^2 = 2e^{-x}(x+1)^4$ ✓(method)

$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = 2\left(-e^{-x}(x+1)^4 + e^{-x} \times 4(x+1)^3\right)$

$\therefore \ddot{x} = 2e^{-x}(x+1)^3[4 - (x+1)]$

$= 2e^{-x}(x+1)^3(3 - x)$

(ii) Note that $v > 0\ \forall x$. Initially, $v = 2 \times e^0(0+1)^2 = 2\text{ ms}^{-1}$, so particle always has positive displacement after $t = 0$. $\ddot{x} = 0$ when $x = 3$. Also, $\ddot{x} > 0$ for $0 < x < 3$ and $\ddot{x} < 0$ for $x > 3$. $\therefore$ Max. velocity occurs when $x = 3$.

Note: $\ddot{x} = 0$ when $x = 3$ is not sufficient justification, as it ignores the possibility of the object speeding up for $x > 3$ (i.e. v having an inflection at $x = 3$).

HSC Extension 2 Mathematics

428

Mechanics Solution

The largest acceleration would be produced by the greatest force.
We need F to be as large as possible with θ closest to 180° if possible.

So C is eliminated in favour of B

Even though option A has 180° with it $\left|2\sqrt{3} \times -\frac{\sqrt{3}}{2}\right| = 3$.

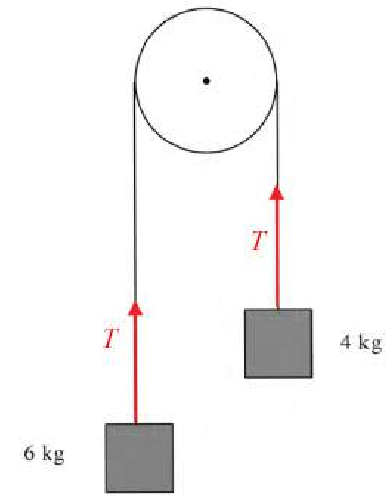
So A is eliminated.

With option D we get $\left|5 \times -\frac{1}{2}\right| = 2.5$.

HSC Extension 2 Mathematics

427

Mechanics Solution



Let a = acceleration of the system

$4a = T - 4g \quad -(1)$

$6a = 6g + T \quad -(2)$

$(1) + (2)$: $10a = 2g$

$a = \frac{g}{5}$

HSC Extension 2 Mathematics

429

Mechanics

A particle moves in a straight line so that at time t its displacement from a fixed origin is x and its velocity is v. **3**

If $\ddot{x} = -2e^{-x}$ and $v = 2$, $x = 0$ when $t = 0$, find x as a function of t.

HSC Extension 2 Mathematics

430

Mechanics

The displacement of a particle is given by $x = \sqrt{3}\sin 2t - \cos 2t + 3$.

(i) Find the initial displacement and velocity of the particle. **2**

(ii) Prove that the particle is moving in simple harmonic motion. **1**

(iii) Find v^2 in terms of x. **2**

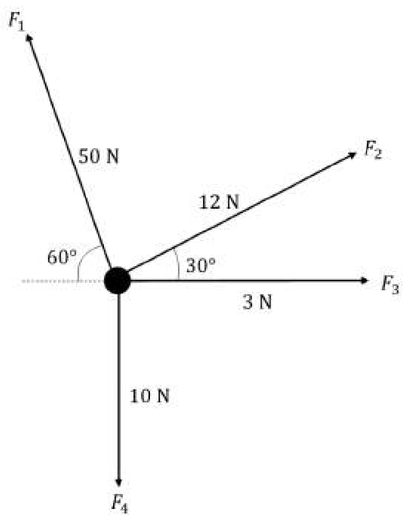
(iv) Hence or otherwise find the maximum displacement and maximum speed of the particle. **2**

HSC Extension 2 Mathematics

431

Mechanics

Four forces, F_1, F_2, F_3 and F_4 act on a particle as shown. **4**



Find the magnitude and direction of the net force acting on the particle. Leave your answer correct to two decimal places.

HSC Extension 2 Mathematics

432

Mechanics

Hooke's law states that when a light elastic string is stretched out and released, the acceleration on the object attached to the string satisfies:

$$\ddot{x} = -\frac{g}{l}x$$

where $\ddot{x}$ is the acceleration of the object in m s^{-2}, x is the displacement of the object from the starting point in metres, and g and l are constants.

Initially, the object is stretched to a maximum displacement of $\frac{l}{2}$ metres from the origin and is at rest.

(i) Show that: **3**

$$v = \pm\sqrt{\frac{g}{l}} \times \sqrt{\frac{l^2}{4} - x^2}$$

(ii) Find the period and amplitude of motion. **2**

(iii) Describe what happens to the velocity and acceleration as $l \to \infty$. **1**

HSC Extension 2 Mathematics

430

Mechanics Solution

(i)

when $t = 0$

$$x = \sqrt{3}\sin(2(0)) - \cos(2(0)) + 3$$
$$= 0 - 1 + 3 = 2\text{ m}$$

$$\dot{x} = 2\sqrt{3}\cos 2t + 2\sin 2t$$

when $t = 0$

$$\dot{x} = 2\sqrt{3}\cos(2(0)) + 2\sin(2(0))$$
$$= 2\sqrt{3}\text{ m/s}$$

(ii)

$$\ddot{x} = -4\sqrt{3}\sin 2t + 4\cos 2t$$
$$= -4(\sqrt{3}\sin 2t - \cos 2t)$$
$$= -4(x-3)$$

$\therefore$ particle is moving in simple harmonic motion as acceleration is proportional to displacement in the opposide direction

(iii)

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = -4(x-3)$$

$$\frac{d}{dx}v^2 = -8(x-3)$$

$$\left[v^2\right]_{2\sqrt{3}}^{v} = \int_2^x -8x + 24\, dx$$

$$v^2 - (2\sqrt{3})^2 = \left[-4x^2 + 24x\right]_2^x$$

$$v^2 - 12 = -4x^2 + 24x - (-4\times 2^2 + 24(2))$$
$$= -4x^2 + 24x - 32$$

$$\therefore v^2 = -4x^2 + 24x - 20$$

(iv)

$$v^2 = -4(x^2 - 6x + 5)$$
$$= -4(x-5)(x-1)$$

$\therefore v^2 = 0$ at $x = 5$ & $x = 1$

$\therefore$ max displacement at $x = 5$

v^2 is maximised at $\frac{5+1}{2} = 3$

$\therefore v^2 = -4(3-5)(3-1) = 16$

$\therefore$ max speed $= 4$ m/s

HSC Extension 2 Mathematics

429

Mechanics Solution

$$\ddot{x} = -2e^{-x}$$

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = -2e^{-x}$$

$$\frac{1}{2}v^2 = 2\int(-e^{-x})\,dx$$

$$\frac{1}{2}v^2 = 2e^{-x} + c_1$$

$x = 0,\ v = 2 \implies c_1 = 0$

$$\therefore v^2 = 4/e^x$$

$$v = 2e^{-x/2} \qquad v = \frac{dx}{dt}$$

$$\frac{dx}{dt} = 2e^{-x/2}$$

separating the variables

$$\frac{dt}{dx} = \frac{1}{2}e^{x/2}$$

$$\therefore t = e^{x/2} + c_2$$

$t = 0,\ x = 0, \implies c_2 = -1$

$$e^{x/2} = 1 + t, \qquad x/2 = \ln|1+t|$$

$$\therefore x = 2\ln|1+t|$$

HSC Extension 2 Mathematics

432

Mechanics Solution

(i)
$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = -\frac{g}{l}x$$

$$\frac{v^2}{2} = -\frac{gx^2}{2l} + C$$

When $v = 0, x = \frac{l}{2}$:

$$C = \frac{g\left(\frac{l}{2}\right)^2}{2l} = \frac{gl}{8}$$

$$\frac{v^2}{2} = -\frac{gx^2}{2l} + \frac{gl}{8}$$

$$v^2 = -\frac{gx^2}{l} + \frac{gl}{4}$$

$$v^2 = \frac{g}{l}\left(\frac{l^2}{4} - x^2\right)$$

$$v = \pm\sqrt{\frac{g}{l}} \times \sqrt{\frac{l^2}{4} - x^2}$$

(ii)
$$T = \frac{2\pi}{n} = \frac{2\pi}{\sqrt{\frac{g}{l}}} = 2\pi\sqrt{\frac{l}{g}}$$

$$a = \sqrt{\frac{l^2}{4}} = \frac{l}{2}$$

(iii) As $l \to \infty, v \to \infty$ and $a \to 0$.

HSC Extension 2 Mathematics

431

Mechanics Solution

Take the upward and right direction to be positive and the downward and left direction to be negative.

Horizontally:

$$F_{horizontal} = 3 + 12\cos 30 - 50\cos 60$$
$$= 3 + 6\sqrt{3} - 25$$
$$= -22 + 6\sqrt{3}$$

Vertically:

$$F_{vertical} = 50\sin 60 + 12\sin 30 - 10$$
$$= 25\sqrt{3} + 6 - 10$$
$$= 25\sqrt{3} - 4$$

$$F_{net} = \sqrt{(-22 + 6\sqrt{3})^2 + (25\sqrt{3} - 4)^2} = 40.98\text{ N}$$

$$\theta = 180° - \tan^{-1}\left(\frac{25\sqrt{3} - 4}{-22 + 6\sqrt{3}}\right)$$

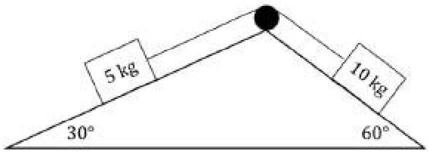
$$\theta = 106.45°$$

HSC Extension 2 Mathematics

433

Mechanics

The diagram shows two masses on either side of a slope, connected by a light inextensible string that runs through a smooth pulley.



The 5 kg mass is on a slope that is inclined at an angle of 30° to the horizontal. The surface of this slope is rough and produces a frictional force on the 5 kg mass equal to μN, where μ is a constant and N is the normal force of the 5 kg mass.

The 10 kg mass is on a slope that is inclined at 60° to the horizontal. The surface of this slope is frictionless.

Let the acceleration due to gravity is g m s^{-2} and T be the tension force in the string (measured in Newtons).

(i) Show that the acceleration of the 5 kg mass up the slope is given by: **3**

$$a = \frac{1}{5}\left(T - \frac{5g}{2} - \frac{5\mu g\sqrt{3}}{2}\right)$$

(ii) It is given that the acceleration the acceleration of the 10 kg mass down the slope is:

$$a = \frac{g\sqrt{3}}{2} - \frac{T}{10}$$

If both masses are moving at a constant speed along their respective slopes, show that: **3**

$$\mu = \frac{1}{3}(6 - \sqrt{3})$$

HSC Extension 2 Mathematics

434

Mechanics

A particle is moving along the x axis.

Its acceleration is given by $a = \dfrac{7-3x}{x^3}$, and the particle starts from rest at the point $x = 1$.

(i) Explain why the particle starts moving in the positive x direction. **1**

(ii) Let v be the velocity of the particle. Show that $v = \dfrac{\sqrt{x^2+6x-7}}{x}$. **3**

(iii) Describe the motion of the particle after it passes $x = \dfrac{7}{3}$. **1**

HSC Extension 2 Mathematics

435

Mechanics

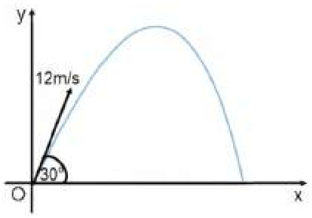
A projectile is fired from the origin O with initial velocity 12 ms^{-1} at an angle of 30° to the horizontal and experiences air resistance proportional to the velocity in both the x and y directions. The equations of motion are given by

$$\ddot{x} = -k\dot{x} \qquad \ddot{y} = -10 - k\dot{y}$$

$$\dot{x} = 6\sqrt{3}e^{-kt} \qquad \dot{y} = \frac{1}{k}\left[(10+6k)e^{-kt} - 10\right]$$

$$x = \frac{6\sqrt{3}}{k}(1 - e^{-kt}) \qquad y = \frac{10+6k}{k^2}(1 - e^{-kt}) - \frac{10}{k}t$$

where k is a positive constant. (Do NOT prove these results.)



(i) Show that the Cartesian equation of the flight path is given by **2**

$$y = \frac{5+3k}{3k\sqrt{3}}x + \frac{10}{k^2}\ln\left(1 - \frac{kx}{6\sqrt{3}}\right)$$

(ii) Given that $k = 0.1$, find the maximum height reached in metres to 2 decimal places. **3**

HSC Extension 2 Mathematics

436

Mechanics

A ball of mass 2 kilograms is thrown vertically upward from the origin with an initial speed of 8 metres per second. The ball is subject to a downward gravitational force of 20 newtons and an air resistance of $\dfrac{v^2}{5}$ newtons in the opposite direction to the velocity, v metres per second.

Hence, until the ball reaches its highest point, the equation of motion is

$$\ddot{y} = -\frac{v^2}{10} - 10, \qquad \text{(Do NOT prove this.)}$$

where y metres is its height.

(i) Show that, while the ball is rising, $v^2 = 164e^{-\frac{y}{5}} - 100$. **2**

(ii) Hence find the maximum height reached. **1**

(iii) Find how long the ball takes to reach this maximum height. **2**

(iv) Write down the equation of motion for the downwards journey. **1**

(v) How fast is the ball travelling when it returns to the origin? **2**

Mechanics Solution

(i) $t=0,\ x=1,\ \dot{x}=0$ (given), $\ddot{x}=\frac{7-3}{1^3}>0$

Starting from rest and accelerating to right

$\therefore$ initially moves in +ve direction

(ii) $\frac{d}{dx}\left(\frac{1}{2}v^2\right)=\frac{7-3x}{x^3}=7x^{-3}-3x^{-2}$

$$\left[\tfrac{1}{2}v^2\right]_0^v=\int_1^x\left(7x^{-3}-3x^{-2}\right)dx$$

$$\left[v^2\right]_0^v=2\left[\frac{3}{x}-\frac{7}{2x^2}\right]_1^x$$

$$v^2=2\left(\frac{3}{x}-\frac{7}{2x^2}-3+\frac{7}{2}\right)$$

$$=\frac{6}{x}-\frac{7}{x^2}+1$$

$$=\frac{x^2+6x-7}{x^2}$$

$$v=\pm\frac{\sqrt{x^2+6x-7}}{x}$$

at $x^2+6x-7=0 \Rightarrow x=-7,1$, so once particle move to right of $x=1$ it can't stop

$\therefore v>0\ \forall t>0$, $\therefore v=\frac{\sqrt{x^2+6x-7}}{x}$ [3]

(iii) (I) $v>0$ for all time, and as $x\to\infty$, $v\to 1^-$

(II) if $x>\frac{7}{3}$, $a<0$

(III) when $x=7/3$, $v=\frac{4}{\sqrt{3}}$

$\therefore$ particle moves forever to the right, slowing from $v=\frac{4}{\sqrt{3}}$ to a limiting velocity of 1m/s.

Mechanics Solution

(i) For the 5 kg mass:

Net Force = Tension – Force down the slope – Friction

$$ma=T-F_{down}$$

$$5a=T-5g\sin 30°-5\mu g\cos 30°$$

$$5a=T-\frac{5g}{2}-\frac{5\mu g\sqrt{3}}{2}$$

$$a=\frac{1}{5}\left(T-\frac{5g}{2}-\frac{5\mu g\sqrt{3}}{2}\right)$$

(ii) Since the system moves at a constant velocity, $a=0$. Hence from (i):

$$T=\frac{5g}{2}+\frac{5\mu g\sqrt{3}}{2}\ \ldots(1)$$

and from the given equation:

$$T=5g\sqrt{3}\ \ldots(2)$$

Equating (1) and (2):

$$\frac{5g}{2}+\frac{5\mu g\sqrt{3}}{2}=5g\sqrt{3}$$

$$g+\mu g\sqrt{3}=2g\sqrt{3}$$

$$\mu=\frac{2g\sqrt{3}-g}{g\sqrt{3}}$$

$$\mu=\frac{2\sqrt{3}-1}{\sqrt{3}}=\frac{6-\sqrt{3}}{3}$$

Mechanics Solution

(i)

$$\ddot{y}=v\frac{dv}{dy}$$

$$\therefore v\frac{dv}{dy}=-10-\frac{v^2}{10}$$

$$\frac{dv}{dy}=\frac{-10}{v}-\frac{v}{10}$$

$$\frac{dy}{dv}=\frac{-10v}{v^2+100}$$

$$\therefore [y]_0^y=\int_8^v\frac{-10v}{v^2+100}dv$$

$$y=-5\left[\ln(v^2+100)\right]_8^v$$

$$-\frac{y}{5}=\ln(v^2+100)-\ln 164$$

$$-\frac{y}{5}=\ln\left(\frac{v^2+100}{164}\right)$$

$$\frac{v^2+100}{164}=e^{-\frac{y}{5}}$$

$\therefore v^2=164e^{-\frac{y}{5}}-100$ as required.

(ii) Max height occurs when $v=0$.

$$\therefore 0=164e^{-\frac{y}{5}}-100$$

$$100=164e^{-\frac{y}{5}}$$

$$-5\ln\left(\frac{100}{164}\right)=y$$

$\therefore y=5\ln(1.64)$ m or 2.47m

(iii)

$$\ddot{y}=\frac{dv}{dt}$$

$$\therefore \frac{dv}{dt}=-10-\frac{v^2}{10}$$

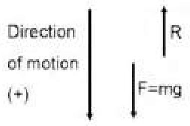
$$\frac{dt}{dv}=\frac{-10}{v^2+100}$$

$$\therefore t=\int_8^0\frac{-10}{v^2+100}dv=-10\times\frac{1}{10}\left[\tan^{-1}\frac{v}{10}\right]_8^0$$

$$=\tan^{-1}\left[\frac{v}{10}\right]_0^8=\tan^{-1}\left(\frac{4}{5}\right)$$

$\therefore$ Reaches max height after $\tan^{-1}\left(\frac{4}{5}\right)$ seconds.

(iv)



$$m\ddot{y}=F-R$$

$$2\ddot{y}=20-\frac{v^2}{5}$$

$$\therefore \ddot{y}=10-\frac{v^2}{10}$$

(v)

$$v\frac{dv}{dy}=10-\frac{v^2}{10}$$

$$\frac{dv}{dy}=\frac{10}{v}-\frac{v}{10}$$

$$\frac{dy}{dv}=\frac{10v}{100-v^2}$$

$$[y]_0^{5\ln 1.64}=\int_0^v\frac{10v}{100-v^2}dv$$

$$5\ln 1.64=-5\left[\ln(100-v^2)\right]_0^v$$

$$-\ln 1.64=\ln(100-v^2)-\ln 100$$

$$\ln(100-v^2)=\ln\left(\frac{100}{1.64}\right)$$

$$v^2=\frac{64}{1.64}$$

$$v=\frac{8}{\sqrt{1.64}}\ (\text{since } v>0)$$

$$=6.25\text{ m/s}$$

$\therefore$ Ball is travelling at 6.25 m/s when it returns to the origin.

Mechanics Solution

(i)

$$\frac{kx}{6\sqrt{3}}=1-e^{-kt}$$

$$e^{-kt}=1-\frac{kx}{6\sqrt{3}}$$

$$-kt=\ln\left(1-\frac{kx}{6\sqrt{3}}\right)$$

$$t=-\frac{1}{k}\ln\left(1-\frac{kx}{6\sqrt{3}}\right)$$

Substitute t into y:

$$y=\frac{10+6k}{k^2}\cdot\frac{kx}{6\sqrt{3}}-\frac{10}{k}\left[-\frac{1}{k}\ln\left(1-\frac{kx}{6\sqrt{3}}\right)\right]$$

$$=\frac{10+6k}{6k^2\sqrt{3}}kx+\frac{10}{k^2}\ln\left(1-\frac{kx}{6\sqrt{3}}\right)$$

$y=\frac{5+3k}{3k\sqrt{3}}x+\frac{10}{k^2}\ln\left(1-\frac{kx}{6\sqrt{3}}\right)$ as required.

(ii) Maximum height occurs when $\dot{y}=0$:

$$0=\frac{1}{k}[(10+6k)e^{-kt}-10]$$

$$(10+6k)e^{-kt}=10$$

$$e^{-kt}=\frac{10}{10+6k}$$

$$e^{kt}=1+0.6k$$

$$kt=\ln(1+0.6k)$$

$$t=\frac{1}{k}\ln(1+0.6k)$$

When $k=0.1$, $t=\frac{1}{0.1}\ln(1+0.6\times 0.1)$

$$t=0.582689\ldots$$

Sub t into y:

$$y=\frac{10+6k}{k^2}(1-e^{-kt})-\frac{10}{k}t$$

$$y=\frac{10+6(0.1)}{0.1^2}(1-e^{-(0.1)(0.582689\ldots)})-\frac{10}{0.1}(0.582689\ldots)$$

$$y=1.73\text{m}$$

HSC Extension 2 Mathematics

437

Mechanics

A particle of mass m is moving in a straight line under the action of an applied force $F = \frac{m}{x^3}(4+6x)$.
What is the equation for its velocity at any position if the particle starts from rest at $x = 1$?

A. $v = \pm\frac{1}{x}\sqrt{8x^2-6x-2}$

B. $v = \pm\frac{2}{x}\sqrt{4x^2-3x-1}$

C. $v = \frac{-4}{x^2}-\frac{12}{x}+16$

D. $v = \sqrt{\frac{-4}{x^2}-\frac{12}{x}}$

HSC Extension 2 Mathematics

438

Mechanics

(i) Prove that a particle moving according to the equation $|v| = \sqrt{-9x^2+12x+32}$ is undergoing simple harmonic motion. **2**

(ii) State the period of motion. **1**

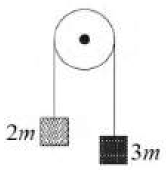
(iii) Find the range of the motion. **1**

HSC Extension 2 Mathematics

439

Mechanics

Two masses, $2m$ kg and $3m$ kg, are connected by a light inextensible string passing over a frictionless pulley as shown. **3**



Initially, the two masses are at rest. After the lighter mass has travelled x metres in an upwards direction, it is travelling at v m s^{-1}.

Prove that $v = \sqrt{\frac{2gx}{5}}$.

HSC Extension 2 Mathematics

440

Mechanics

A particle moves with acceleration function $\ddot{x} = 3x^2$. Initially $x = 1$ and $v = -\sqrt{2}$.

(i) Show that $v^2 = 2x^3$. **2**

(ii) Explain why the velocity can never be positive. **1**

(iii) Find the displacement-time function, and briefly describe the motion. **3**

HSC Extension 2 Mathematics

438

Mechanics Solution

i) $|v| = \sqrt{-9x^2+12x+32}$

$\frac{1}{2}v^2 = -\frac{9}{2}x^2 + 6x + 16$

$\ddot{x} = \frac{d}{dx}\left(-\frac{9}{2}x^2 + 6x + 16\right)$

$= -9x + 6$

$= -9\left(x - \frac{2}{3}\right)$ (in the form $-n^2(x-c)$)

ii) $n^2 = 9$

$n = 3$

Period $= \frac{2\pi}{3}$

iii) $\frac{1}{2}v^2 = -\frac{9}{2}x^2 + 6x + 16$

$v^2 = -9x^2 + 12x + 32$

$= 9\left(-x^2 + \frac{12}{9}x + \frac{32}{9}\right)$

$= 9\left(-x^2 + \frac{4}{3}x - \frac{4}{9} + \frac{32}{9} + \frac{4}{9}\right)$

$= 9\left(-\left(x - \frac{2}{3}\right)^2 + 4\right)$

$= 9\left(4 - \left(x - \frac{2}{3}\right)^2\right)$

$\therefore a^2 = 4, \ a = 2.$

range: ± 2

2 units away from the centre of motion.

$\left(\frac{8}{3} \text{ and } -\frac{4}{3}\right)$

Centre of motion: $\frac{+2}{3} \pm 2$.

HSC Extension 2 Mathematics

437

Mechanics Solution

$F = ma$

$\ddot{x} = \frac{1}{x^3}(4 + 6x)$

$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = \frac{4}{x^3} + \frac{6}{x^2}$

$\frac{1}{2}v^2 = \int (4x^{-3} + 6x^{-2})\,dx + c$

$\frac{1}{2}v^2 = \frac{4x^{-2}}{-2} + \frac{6x^{-1}}{-1} + c$

at $t = 0$, $x = 1$, $v = 0$

$0 = -2 - 6 + c$

$c = 8$

$\frac{1}{2}v^2 = -2x^{-2} - 6x^{-1} + 8$

$\Rightarrow v^2 = -4x^{-2} - 12x^{-1} + 16$

$v^2 = \frac{16x^2 - 12x - 4}{x^2}$

$v = \pm\sqrt{\frac{16x^2 - 12x - 4}{x^2}}$

$v = \pm \frac{2}{x}\sqrt{4x^2 - 3x - 1}$

$\boxed{B}$

HSC Extension 2 Mathematics

440

Mechanics Solution

(a) Since acceleration is given as a function of displacement, we write:

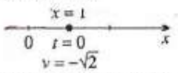
$$\frac{d}{dx}\left(\tfrac{1}{2}v^2\right) = 3x^2$$

$$\tfrac{1}{2}v^2 = x^3 + \tfrac{1}{2}C, \quad \text{for some constant } C.$$

$$v^2 = 2x^3 + C, \quad \text{for some constant } C.$$

When $x = 1$, $v = -\sqrt{2}$, so $\quad 2 = 2 + C,$

so $C = 0$, and $\quad v^2 = 2x^3.$



(b) Taking square roots, $\quad v = -\sqrt{2}\,x^{\frac{3}{2}}$, assuming that v is never positive.

Taking reciprocals, $\quad \frac{dt}{dx} = -\tfrac{1}{2}\sqrt{2}\,x^{-\frac{3}{2}}$

$$t = \sqrt{2}\,x^{-\frac{1}{2}} + D, \quad \text{for some constant } D.$$

When $t = 0$, $x = 1$, so $\quad 0 = \sqrt{2} + D,$

so $D = -\sqrt{2}$, and $\quad t = \sqrt{2}\,x^{-\frac{1}{2}} - \sqrt{2}$

$$x^{-\frac{1}{2}} = \frac{t + \sqrt{2}}{\sqrt{2}}$$

$$x = \frac{2}{(t + \sqrt{2})^2}.$$

Hence the particle begins at $x = 1$, and moves backwards towards the origin. As $t \to \infty$, its speed has limit zero, and its limiting position is $x = 0$.

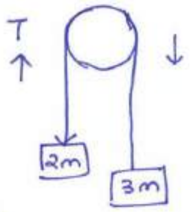
(c) Initially, v is negative. Since $v^2 = 2x^3$, it follows that v can only be zero at the origin; but since $\ddot{x} = 3x^2$ the acceleration at the origin would also be zero. Hence if the particle ever arrived at the origin it would then be permanently at rest. Thus the velocity can never change from negative to positive.

HSC Extension 2 Mathematics

439

Mechanics Solution

Net force :



$T - 2mg = 2m\ddot{x}$

Also, $\quad 3mg - T = 3m\ddot{x}$

adding the two equations

$mg = 5m\ddot{x}$

$\ddot{x} = \frac{g}{5}$

$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = \frac{g}{5}$

$\frac{1}{2}v^2 = \frac{gx}{5} + c$

at $x = 0$, $v = 0 \Rightarrow c = 0$

$\frac{1}{2}v^2 = \frac{gx}{5}$

$v^2 = \frac{2gx}{5}$

$v = \sqrt{\frac{2gx}{5}} \quad (v > 0)$